

Geometry Crystallisation in CRF: v4

Two Regimes, the Absolute- N_{fire} Control Variable, and Estimator Bias at
Large N

Pattern analysis across $N \in \{125, 250, 400, 500, 1000\}$ with corrected honest boundaries

Ougit Jarittum

ORCID: 0009-0009-4451-6811

April 2026

Abstract

Pattern analysis across five system sizes ($N \in \{125, 250, 400, 500, 1000\}$, including 1000-sweep verification at $N = 500$) reveals three results not present in v1–v3.

(1) N_{fire} (absolute) is the control variable, not $N_{\text{fire}}/N_{\text{crit}}$. $\text{Corr}(N_{\text{fire}}, \langle d_s \rangle) = 0.88$; $d_s = 3$ when $N_{\text{fire}} \approx 11$ events per sweep. Linear fit: $\langle d_s \rangle = 0.167 N_{\text{fire}} + 1.16$. The inverse correlation $\text{Corr}(N_{\text{fire}}/N_{\text{crit}}, \langle d_s \rangle) = -0.97$ was misleading — the ratio decreases monotonically with N while d_s increases, because both are driven by N in opposite directions.

(2) Two-regime σ structure. $\sigma(d_s) \approx 0.64$ (constant) for $N < N^*$ (pre-threshold regime); $\sigma(d_s) \approx 0.36$ for $N \geq N^*$ (3D-locked shot-noise regime). The ratio $0.64/0.36 \approx \sqrt{3} = 1.73$ (gap 3%). The pre-threshold σ is threshold noise (system oscillates between $d_s \approx 1$ and $d_s \approx 4$); the post-threshold σ is shot noise from concurrent firing.

(3) $N = 1000$ overshoot is estimator bias, not burn-in. $\tau(N) = f_{\text{II}}/(R/N) \approx 8$ sweeps (constant, N -independent), ruling out burn-in as the source. The fixed walk-times $[2, 4, \dots, 18]$ under-probe the return-probability slope at $N = 1000$ ($N_{\text{fire}} \approx 16$), producing an upward bias in d_s . Corrected estimator (walk-times scaled with N) is needed for $N > 500$.

Contents

1	Data: Multi-N Scaling Results	2
2	Pattern 1: Absolute N_{fire} as Control Variable	2
2.1	The True Predictor	2
2.2	Why the Ratio Was Misleading	2
2.3	Physical Interpretation	2
3	Pattern 2: Two-Regime σ Structure	3
4	Pattern 3: $N = 1000$ Bias and τ Scaling	3
4.1	$\tau(N)$ is Constant	3

4.2	Estimator Bias at Large N	4
4.3	Scale Separation Confirmed	4
5	Status Table	4
6	Open Questions	5
7	Key Numbers	5

1 Data: Multi- N Scaling Results

N	Sweeps	$\langle d_s \rangle$	$\sigma(d_s)$	N_{fire}	N_{crit}	Regime
125	400	1.087	0.650	3.2	18	Pre-threshold
250	500	2.078	0.655	5.0	37	Pre-threshold
400	600	2.716	0.636	7.0	59	Approaching
500	1000	2.994	0.358	8.4	73	3D-locked
1000	600	3.526	0.370	16.0	147	Biased (see §4.2)

Note: $N_{\text{crit}} = \lfloor f_{\text{II,crit}} \cdot N \rfloor$ where $f_{\text{II,crit}} = 0.147$ (derived, gap 0.004%). N_{fire} is the mean R-events per sweep in the late run. $R/N \approx 0.019$ is approximately constant across all N .

2 Pattern 1: Absolute N_{fire} as Control Variable

2.1 The True Predictor

N_{fire} controls $\langle d_s \rangle$

$$\langle d_s \rangle = 0.167 N_{\text{fire}} + 1.16, \quad \text{Corr}(N_{\text{fire}}, \langle d_s \rangle) = 0.88$$

$d_s = 3$ when $N_{\text{fire}} \approx 11$ events per sweep. This is a *threshold on the absolute firing rate*, not on the ratio $N_{\text{fire}}/N_{\text{crit}}$.

2.2 Why the Ratio Was Misleading

The earlier finding $\text{Corr}(N_{\text{fire}}/N_{\text{crit}}, \langle d_s \rangle) = -0.97$ appeared to identify $N_{\text{fire}}/N_{\text{crit}}$ as the universal control. The inverse correlation arose because:

- $N_{\text{crit}} = f_{\text{II,crit}} \cdot N$ grows with N ;
- $N_{\text{fire}} \approx 0.019 N$ also grows with N , but slightly slower;
- Therefore $N_{\text{fire}}/N_{\text{crit}} \approx 0.019/0.147 \approx 0.13$ decreases slightly as N grows due to finite- N corrections;
- $\langle d_s \rangle$ increases with N .

Both ratios are monotone functions of N , creating a spurious inverse correlation. The actual mechanism is: larger N produces more concurrent firers, which provide a stronger 3D consensus signal per sweep.

2.3 Physical Interpretation

The critical condition for 3D lock is not “ N_{fire} exceeds N_{crit} per sweep” (since $N_{\text{fire}} \ll N_{\text{crit}}$ at all N), but rather: “ N_{fire} provides enough 3D-biased votes to overcome the isotropic

noise from Phase I firers.”

From the linear fit: each additional R-event per sweep contributes 0.167 units to $\langle d_s \rangle$. The threshold at $N_{\text{fire}} = 11$ is the absolute firing rate at which Phase II nodes dominate the geometric vote.

3 Pattern 2: Two-Regime σ Structure

Two-Regime σ Transition

$$\sigma(d_s) \approx \begin{cases} 0.64 & N < N^* \approx 500 \quad (\text{pre-threshold, constant}) \\ d_s \sqrt{f_{\text{II}}(1 - f_{\text{II}})/N_{\text{fire}}} \approx 0.36 & N \geq N^* \quad (\text{shot-noise, } \propto N^{-0.5}) \end{cases}$$

The ratio at the transition: $\sigma_{\text{pre}}/\sigma_{\text{post}} = 0.64/0.36 = 1.78 \approx \sqrt{3} = 1.732$ (gap 3%).

Pre-threshold regime ($N < N^*$). When $N_{\text{fire}} < 11$ per sweep, the geometry cannot sustain 3D consensus. d_s oscillates between ~ 1 (Phase I dominated) and ~ 4 (Phase II dominated but overshoot), producing $\sigma \approx 0.64$ that is independent of N . This is *threshold noise* — the fluctuation of a system that is bistable between the pre-geometric and 3D-locked attractor.

Post-threshold regime ($N \geq N^*$). When $N_{\text{fire}} \geq 11$, Phase II nodes dominate the vote and $d_s \approx 3$. The residual fluctuation is shot noise from the discrete firing process: $\sigma = d_s \sqrt{f_{\text{II}}(1 - f_{\text{II}})/N_{\text{fire}}}$, which decreases as $N_{\text{fire}} \propto N$ grows.

The $\sqrt{3}$ ratio. $\sigma_{\text{pre}}/\sigma_{\text{post}} = 0.64/0.36 = 1.78 \approx \sqrt{d_s} = \sqrt{3}$. Interpretation: the threshold-noise amplitude equals $\sqrt{d_s}$ times the post-threshold shot-noise amplitude. Each spatial dimension contributes an equal amount of geometric variance at the threshold, and the total threshold noise is the geometric mean of the d_s shot-noise modes. This connection requires formal derivation (Open, §6).

4 Pattern 3: $N = 1000$ Bias and τ Scaling

4.1 $\tau(N)$ is Constant

Burn-in Time is N -Independent

$$\tau(N) \approx \frac{f_{\text{II,crit}}}{R/N} = \frac{0.147}{0.019} \approx 8 \text{ sweeps}$$

Since $R/N \approx 0.019$ is constant (confirmed across $N = 125$ – 500), τ is N -independent. The $N = 1000$ overshoot ($\langle d_s \rangle = 3.53$) cannot be attributed to insufficient burn-in.

4.2 Estimator Bias at Large N

The d_s estimator uses a fixed random-walk schedule: `walk_times` = [2, 4, 6, ..., 18] steps. The return-probability scaling $P(t) \propto t^{-d_s/2}$ is extracted from a log-linear fit over this range.

The characteristic network distance (mean shortest path between nodes) scales as $\ell \propto \sqrt{N}$ for the k -nearest-neighbour graph: $\ell(N = 500) \approx 22$ hops, $\ell(N = 1000) \approx 32$ hops.

Bias Mechanism

At $N = 500$: walk-time 18 covers $18/22 \approx 0.8$ mean network distances — the slope is measured over the full scaling regime.

At $N = 1000$: walk-time 18 covers $18/32 \approx 0.6$ mean network distances — walkers have not yet reached the diffusive regime. Return probabilities are overestimated (walkers haven't diffused far), producing an upward bias in d_s .

Fix: scale walk-times with N : `walk_times` = [2, 4, ..., $\lceil 18 \cdot (N/500)^{0.5} \rceil$]. At $N = 1000$: max walk-time $\rightarrow 25$ steps.

4.3 Scale Separation Confirmed

The ϵ_0 vs $\sigma(d_s)$ separation established in v3 holds: $\epsilon_0 = 0.00755$ (fixed, microscopic) and $\sigma \approx 0.36$ (macroscopic, N -dependent) differ by $47\times$. In the thermodynamic limit $N \rightarrow \infty$: $\sigma \rightarrow 0$, ϵ_0 fixed.

5 Status Table

Result	Status	Note
N_{fire} controls $\langle d_s \rangle$	Derived	linear fit, corr 0.88
$d_s = 3$ at $N_{\text{fire}} \approx 11/\text{sw}$	Derived	from linear fit
Two-regime σ (0.64 / 0.36)	Confirmed	ratio $\approx \sqrt{3}$
$\sigma_{\text{pre}}/\sigma_{\text{post}} = \sqrt{d_s}$	Hypothesis	gap 3%
$\tau(N) = f_{\text{II}}/(R/N) \approx 8 \text{ sw}$	Derived	N -independent
$N = 1000$ overshoot = estimator bias	Derived	walk-time too short
Scale separation ϵ_0 vs σ	Confirmed	$47\times$
$N_{\text{fire}}/N_{\text{crit}}$ inverse corr.	Artifact	spurious, both $\propto N$
$\sigma \sim N^{-0.5}$ (shot-noise)	Hypothesis	valid for $N \geq N^*$ only
$\sigma_{\text{pre}} = \sqrt{d_s} \cdot \sigma_{\text{post}}$ algebraic	Open	§6

6 Open Questions

Why $\sigma_{\text{pre}}/\sigma_{\text{post}} = \sqrt{d_s}$?

The ratio $0.64/0.36 = 1.78 \approx \sqrt{3}$ is intriguing. Formal derivation: show that the threshold noise amplitude equals $\sqrt{d_s}$ times the post-threshold shot noise. Physical route: at threshold, each of the d_s spatial axes independently contributes σ_{post} of geometric variance, giving total amplitude $\sqrt{d_s} \cdot \sigma_{\text{post}}$.

Time normalization collapse (killer test)

After correcting the estimator at $N = 1000$, plot $d_s(t/\tau(N))$ for all N on the same axes. Since $\tau(N) \approx 8$ sw (constant), the plot is simply d_s vs t (unrescaled), but with $N = 1000$ corrected. If curves collapse: confirms universal dynamics. If not: separate hidden variable beyond N_{fire} .

$N_{\text{fire}} = 11$ threshold from δ -chain

The linear fit gives $d_s = 3$ at $N_{\text{fire}} \approx 11$. CRF prediction route: $N_{\text{fire}}^* = f_{\text{II,crit}} \cdot N^* \cdot (R/N)/f_{\text{II,crit}} = N^* \cdot (R/N)$. With $R/N \approx 0.019$ and $N^* \approx 500$: $N_{\text{fire}}^* \approx 9.5$. Gap from 11: 16%. Closing this requires deriving R/N from T_{avg} and the threshold condition from the $\text{SO}(3)$ percolation argument.

7 Key Numbers

$\langle d_s \rangle = 0.167 N_{\text{fire}} + 1.16$	$\text{Corr}(N_{\text{fire}}, d_s) = 0.88$
$d_s = 3$ at $N_{\text{fire}} \approx 11/\text{sw}$	$R/N \approx 0.019$ (const.)
$\sigma_{\text{pre}} \approx 0.64$ (const.)	$\sigma_{\text{post}} \approx 0.36$ ($N = 500$)
Ratio $\sigma_{\text{pre}}/\sigma_{\text{post}} = 1.78 \approx \sqrt{3}$	HN-04 violations = 0 (all N)
$\tau(N) \approx 8$ sw (const.)	Estimator bias at $N > 500$: fix walk-times

The geometry does not ask how many nodes exist.

It asks how many fired this sweep.

Eleven concurrent firers build three dimensions.

Below eleven, the vote is too close to call.

*Above eleven, the consensus grows — and the noise shrinks
as $1/\sqrt{N_{\text{fire}}}$, one vote at a time.*

At the threshold: $\sigma_{\text{pre}} = \sqrt{d_s} \cdot \sigma_{\text{post}}$.

Three dimensions appear exactly where space pays for itself.